# Exercise 6: Library Management System

## 1. Aim

To develop a **Library Management System** using Java and implement **Linear Search** and **Binary Search** algorithms to search books by title.

# 2. Problem Understanding

A library management system stores information about books such as Book ID, Title, and Author. Users frequently search for books using their titles.

Choosing an appropriate search algorithm improves the speed of finding books, especially when the number of books is very large.

# 3. Search Algorithms

## Linear Search

Linear Search checks each book one by one until the required title is found.

**Characteristics:**

- Works on both sorted and unsorted data.
- Easy to implement.
- Suitable for small datasets.

## Binary Search

Binary Search repeatedly divides the sorted array into two halves and compares the middle element with the search key.

**Characteristics:**

- Requires the array to be sorted.
- Much faster for large datasets.
- Efficient due to divide-and-conquer.

# 4. Algorithm

### Linear Search

1. Start from the first book.
2. Compare the title with the search title.

3. If matched, return the book.
4. Otherwise, continue until the end of the array.

**Binary Search**

1. Sort the books by title.
2. Find the middle element.
3. Compare its title with the search title.
4. If equal, return the book.
5. If the title is smaller, search the left half.
6. Otherwise, search the right half.
7. Repeat until the book is found or the search space becomes empty.

# 5. Program/Output:

Exercise6.java > Exercise6 > main(String[])

```java
import java.util.Arrays;
import java.util.Comparator;

class Book{
    int bookId;
    String title;
    String author;

    public Book(int bookId,String title,String author){
        this.bookId= bookId;
        this.title = title;
        this.author = author;
    }

    public String toString(){
        return "BookId: "+bookId+", Title:"+title+", Author: "+author;
    }
}
```

```java
public class Exercise6 {
    public static Book BooklinearSearch(Book[] books,String title){
        for(Book book:books){
            if(book.title.equalsIgnoreCase(title)){
                return book;
            }
        }
        return null;
    }
```

```java
    public static Book BookBinarySearch(Book[]books,String title){

        int low =0;
        int high = books.length-1;

        while(low<=high){
            int mid= (low+high)/2;
            int comparision = books[mid].title.compareToIgnoreCase(title);
            if(comparision==0){
                return books[mid];
            }
            else if(comparision<0){
                low = mid +1;
            }
            else{
                high = mid-1;
            }
        }
        return null;
    }
```

```java
    public static void main(String[] args) {
        Book[] books={
            new Book(bookId: 101, title: "Digital Electronics", author: "V.Vijyanendran"),
            new Book(bookId: 103, title: "Java Programming", author: "James Gosling"),
            new Book(bookId: 104, title: "Data Structure", author: "Mark Allen"),
            new Book(bookId: 102, title: "Computer Network", author: "Galvin"),
        };
        String searchTitle = "Operating System";
        System.out.println(x: "Linear Search");
        Book result1 = BooklinearSearch(books, searchTitle);

        if(result1!=null){
            System.out.println(result1);
        }
        else{
            System.out.println(x: "Book Not Found");
        }

        Arrays.sort(books,Comparator.comparing(book->book.title));
        System.out.println(x: "\nBinary Search");

        String SearchTitle2 = "Digital Electronics";
        Book result2 = BookBinarySearch(books, SearchTitle2);
        if(result2!=null){
            System.out.println(result2);
        }
        else{
            System.out.println(x: "Book Not Found");
        }
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Krishnavishwa\OneDrive\Documents\Material\cognizant\Algorithm>  & 'C:\Program Files
e\User\workspaceStorage\84f21bc9d5619c8f027c7d00daccf3f2\redhat.java\jdt_ws\Algorithm_cc0d2e2b\
Linear Search
Book Not Found

Binary Search
BookId: 101, Title:Digital Electronics, Author: V.Vijyanendran
PS C:\Users\Krishnavishwa\OneDrive\Documents\Material\cognizant\Algorithm>
```

# 6. Time Complexity Analysis

| Algorithm | Best Case | Average Case | Worst Case |
|---|---|---|---|
| **Linear Search** | **O(1)** | **O(n)** | **O(n)** |
| **Binary Search** | **O(1)** | **O(log n)** | **O(log n)** |

# 7. Comparison

| Linear Search | Binary Search |
|---|---|
| **Works on unsorted data** | **Requires sorted data** |
| **Easy to implement** | **More efficient for large datasets** |
| **Slower for large data** | **Faster due to divide-and-conquer** |
| **Time Complexity: O(n)** | **Time Complexity: O(log n)** |

# 8. Result

Thus, the **Library Management System** was successfully implemented using **Linear Search** and **Binary Search** algorithms. Linear Search is suitable for small or unsorted datasets, while Binary Search is more efficient for large sorted datasets due to its **O(log n)** time complexity. Therefore, Binary Search is generally preferred for searching books in large libraries where the data is maintained in sorted order.